

JOONHYUNG KWAK

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SUMMARY

M.S. candidate in Artificial Intelligence at Sungkyunkwan University, graduating Aug. 2026. Specializing in **LLM inference optimization and causal reasoning systems for mental-health applications**. Demonstrated ability to take research prototypes to production: scaled vLLM inference pipelines on 8K+ user corpora, designed LLM causal generation and evaluation prompts, and built end-to-end crawling-to-insight systems consumed by industry stakeholders.

EDUCATION

M.S. in Artificial Intelligence Sungkyunkwan University Advisor: Prof. JinYeong Bak (Human-Language Intelligence Lab) Thesis: <i>From Post to Temporal Causality: Constructing Temporal Causal Symptom Graphs Using Large Language Models</i>	Mar. 2024 – Aug. 2026 (Expected)
Exchange Student in Computer Science Deggendorf Institute of Technology, Germany Advisor: Prof. Patrick Glauner	Mar. 2022 – Jul. 2022
B.S. in Information and Communication Engineering Inha University Advisor: Prof. YooSung Kim Thesis: Audio Data Pre-processing using Attention-based Dynamic Window	Mar. 2020 – Feb. 2024

TECHNICAL SKILLS

LLM Serving	vLLM (PagedAttention, continuous batching), HuggingFace Transformers, LoRA/PEFT, prompt engineering, LLM-as-a-Judge evaluation
Causal ML	Bootstrapped ATE estimation, confidence interval analysis, temporal causal graph construction
Deep Learning	PyTorch, PyTorch Lightning (DDP multi-GPU), TensorFlow
Data Engineering	Playwright, asyncio, pandas, MySQL, AWS
Backend	REST/WebSocket APIs, multi-agent LLM orchestration, Docker, Git
Languages	Python, C/C++, Korean (native), English (fluent)

RESEARCH EXPERIENCE

Human-Language Intelligence Lab, Sungkyunkwan University <i>Graduate Researcher (Advisor: Prof. JinYeong Bak)</i>	Mar. 2024 – Aug. 2026 (Expected)
- Constructed temporal causal symptom graphs using LLM from 230K+ SNS posts by designing a day-level clustering preprocessing pipeline, replacing a legacy causal-matrix approach that failed on sparse user timelines (80% coverage improvement). - Scaled LLM inference throughput by 6× on an 8K-user corpus using vLLM with PagedAttention and continuous batching, reducing end-to-end labeling time from 9h to 1.5h on 2×A6000. - Designed an LLM-as-a-Judge evaluation framework to compare causal explanations generated by baseline versus Our prompts across four LLM backbones (GPT-4o, Qwen3-8B, Llama3-8B, Gemma-7B)	

- Trained multi-label symptom classifiers with PyTorch Lightning DDP on 2 GPUs, achieving 0.78 macro-F1 on PsySym dataset — a 8-point gain over few-shot GPT-4o baselines.

Biomedical AI Training Program, SNU Bundang Hospital

Sep. 2024 – Oct. 2024

Program Trainee

- Built an ECG/ABP preprocessing pipeline (Pan-Tompkins QRS detection, cubic-spline baseline-wander removal via R-peak-anchored knots, resampling to 100 Hz with per-beat normalization to 128 samples) on 1,500 patient records, with quality-filtering rules rejecting 10% of noisy segments.
- Developed a 1D Inception-CNN in PyTorch fusing waveform features with patient demographics via late concatenation for stroke risk prediction, reporting ROC-AUC of 0.81.
- Fine-tuned Phi-2 (2.7B) on the Asclepius clinical-notes dataset via 4-bit NF4 QLoRA for clinical question answering, deploying a Gradio-based A/B comparison demo against the vanilla Phi-2 baseline.

Data Intelligence Lab, Inha University

Nov. 2022 – Dec. 2023

Undergraduate Researcher

- Improved Korean ASR word-error rate from 12.2% to 8.3% by building a text-normalization pipeline handling numerals, loanwords, and disfluencies before Transformer-based decoding.
- Implemented custom Transformer variants in PyTorch for solar-power forecasting, improving MAPE by 7% over the vanilla baseline.

Nursing Informatics Lab, Inha University

Aug. 2022 – May 2023

Research Assistant

- Extracted clinical keyword trends from 5K EMR notes using N-gram tokenization and word2vec clustering, with the resulting analysis published as a chapter in a book [Health Informatics, 4th edition].
- Fine-tuned a domain-specific BERT for medical NER, achieving F1 of 82.8 on 23 entity types, and deployed the model at Ilsan Hospital for clinical use.

SELECTED PROJECTS

Xueqiu Investor Sentiment Pipeline

Nov. 2025 – Dec. 2025

Independent Project — Hedge Fund Data Engineering

- Built an end-to-end async crawler (Playwright + asyncio) handling dynamic rendering, infinite scroll, and anti-bot detection across 7 market tabs on Xueqiu, collecting 700 posts.
- Integrated Fireworks AI (Llama-3) for CN→EN translation and sentiment classification.
- Delivered structured Markdown analyst reports generated with domain specific prompts, consumed by hedge-fund investment team with English documentation.

Suicide & Self-harm Content Detection on SNS

Sep. 2024 – May 2025

Graduate Research (Clinical Collaboration)

- Designed IRB-compliant data collection protocol; recruited 128 participants from clinical settings and collected 3k real SNS posts with text + image metadata.
- Built a GPT-4o structured-output pipeline with hierarchical category taxonomy (6 categories).

Real-time Multi-Agent LLM Debate System

Mar. 2024

ITRC Project

- Architected a real-time multi-agent orchestration backend streaming LLM-to-LLM debate turns over WebSocket, coordinating OpenAI and Gemini agents with role-conditioned prompts and turn-taking state management.
- Handled concurrent token-level streaming and solved agent interleaving deadlocks by introducing a priority queue arbitrator.

Audio Emotion Classification with Dynamic Attention Windows

Feb. 2023 – Jun. 2023

Undergraduate Thesis — Published at KCC 2023

- Identified that fixed-window audio preprocessing discards 25% of emotion-bearing speech segments by visualizing internal attention maps.
- Designed an attention-driven dynamic windowing method that improved classification accuracy from 78% to 84% on AIHub emotion speech corpus.

DeinHaus 4.0 — Smart-Home Anomaly Detection

Feb. 2022 – Aug. 2022

Industry-linked Project, Deggendorf Institute of Technology

- Analyzed multi-sensor ventilation and activity streams from 5 instrumented aging-in-place households to build automated control.

TEACHING

Teaching Assistant, *Introduction to Artificial Intelligence*, SKKU (Autumn 2024)

Teaching Assistant, National Defense AI Education Program, SKKU (Summer 2024)

Teaching Assistant, AI Special Lecture for Non-major Faculty, SKKU (Summer 2024, Winter 2025)

AWARDS

Research Scholarship for Undergraduate Researchers, Inha University (2023)

Han-eum ICT Contest Award (2021)

PUBLICATIONS

J. Kwak et al., “Dataset refining techniques for improving training efficiency of voice-based situation classification models,” Korea Computer Congress (KCC), Poster, 2023.

J. Kwak et al., “From Post to Temporal Causality: Constructing Symptom Temporal Causal Graphs using Large Language Models,” *in preparation for submission*, 2026.